

GHG Inventory and Net-Zero Roadmap for an SME Textile Manufacturer

Portfolio case study · Harshita Sadhna

Based on a project for a Canadian manufacturer through an SME climate clinic. The company is renamed "Northline Knitwear" and all figures are fictional; methods and assumptions reflect the original work.

The brief

Client

Mid-sized knitting mill, ~35 staff, one site in boundary. Sources yarn from India, China and the US and ships to Canada and the US.

Ask

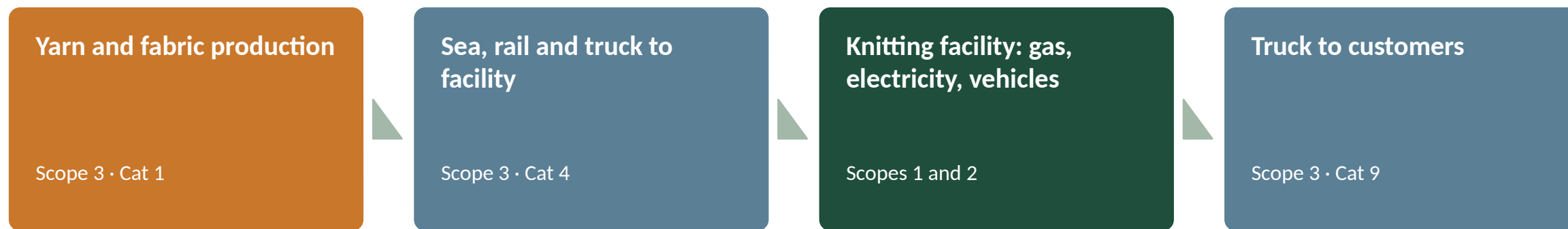
Build a first credible GHG inventory and a practical path to net zero the owners could act on.

My role

Two-person team. I built the calculation model, defined boundaries and assumptions, and drafted the reduction roadmap.

Value chain and what we measured

Operational control approach under the GHG Protocol Corporate Standard



Also included

- Employee commuting (Cat 7), by mode
- Business travel flights (Cat 6)

Out of boundary for now

- Dyeing and finishing site (separate operator)
- Waste, product use and end-of-life (Cats 5, 11, 12)

From spend-based to activity-based

The first pass used dollars spent; the final model uses physical quantities wherever data allowed

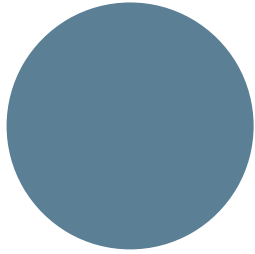
First pass: spend-based

- \$ spent on yarn and freight x industry-average factors
- No split by fibre or sourcing country
- Commuting and travel left out
- Easy to run, but hides where emissions come from

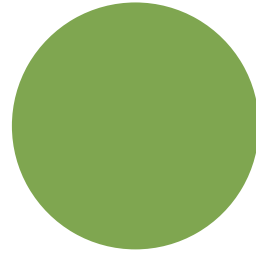
Final model: activity-based

- kg of yarn by fibre and country x LCA factors
- Freight in tonne-km by sea, rail and truck
- Commuting and business travel added
- Several times higher, and shows the hotspots

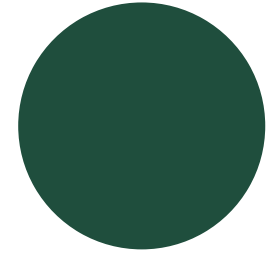
How each number is built



×



=



Activity data

kg, kWh, m³, tonne-km, passenger-km

Emission factor

tCO₂e per unit, from ECCC, US EPA and LCA studies

Emissions

tCO₂e by scope and category

Data sources

Monthly gas and electricity bills · yarn and fabric purchase volumes · freight routes and ports · employee commute survey · flight records. Where data was missing, each gap was filled with a stated assumption so the client could replace it later.

Results: Scope 3 is almost the whole footprint

7,870

tCO₂e total

77

Scope 1

20

Scope 2

98.8%

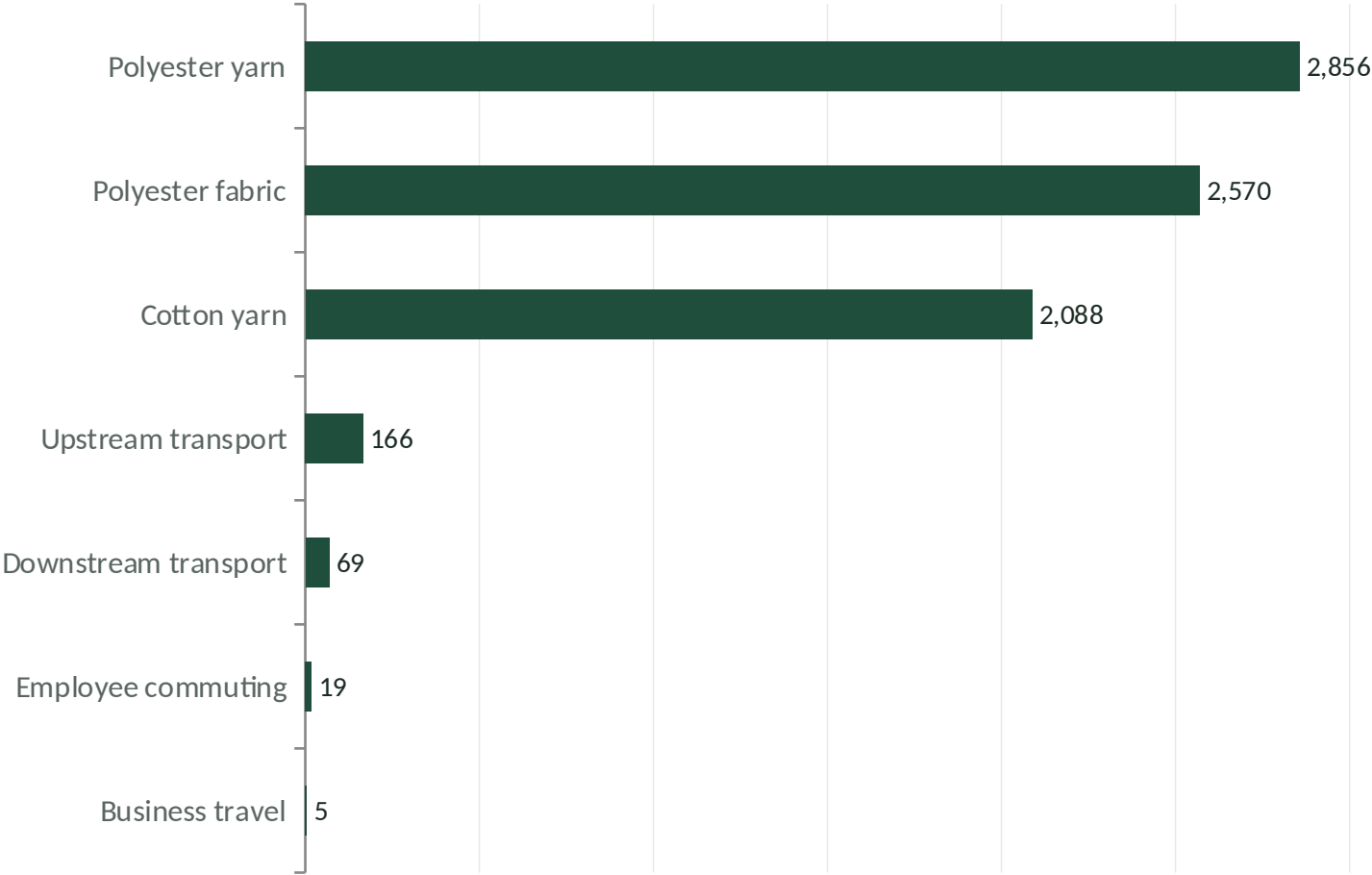
in Scope 3

What this means for the client

- Owned operations are small; Ontario's low-carbon grid keeps Scope 2 near zero.
- Real reductions depend on what yarn is bought and from where, not on factory energy.
- Supplier data is the single biggest lever for accuracy and for cutting emissions.

Purchased materials dominate Scope 3

Scope 3 by category, tCO₂e

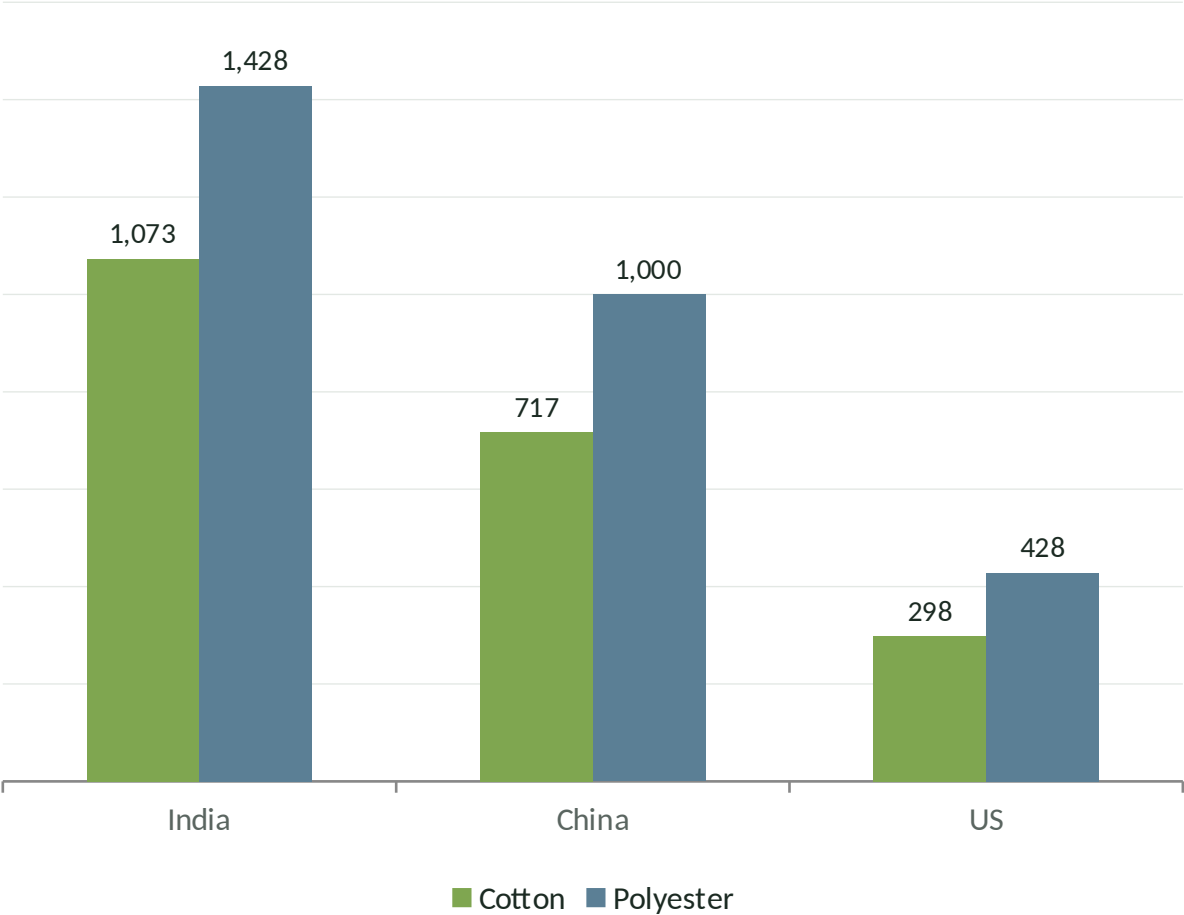


Takeaways

- Yarn and fabric are 95% of Scope 3.
- Polyester carries nearly twice the factor of cotton per kg.
- Freight matters less than expected once measured in tonne-km.

Where yarn comes from changes its footprint

Yarn emissions by sourcing country, tCO₂e



India

Lower cotton yields mean more land, water and fertilizer per kg. Priority for supplier engagement.

China

Higher yields but heavy irrigation and fertilizer. Work with suppliers on inputs.

US

Mechanized, largely rain-fed farming. Lowest factor; shift share here where feasible.

Assumptions made explicit

Every data gap was documented so the estimate can be audited and improved

Sourcing

Volumes split by country and fibre from purchase totals; poly-cotton blends split into fibre shares

Ports and routes

Nearest major export hub per country; sea to West Coast, then rail and truck

Freight loads

15,000 kg per truck to estimate trips; heavy-duty truck factor where fuel data was missing

Customers

Delivery addresses unknown, so regional average distances applied

Fleet

No fuel logs, so a share of freight spend used as a spend-based proxy

Travel

All flights treated as medium-haul, one passenger each; hotels excluded

What the client should collect next year

- | | | |
|---|-------------------------|---|
| 1 | Purchasing | kg of yarn per country and fibre; recycled content share |
| 2 | Inbound freight | Port of origin, shipments per route, carrier fuel or km reports |
| 3 | Outbound freight | Customer city, shipments per customer, weight per trip |
| 4 | Travel | Departure and arrival airports, passengers per trip |
| 5 | New categories | Waste by type, dyeing-site energy use, product disposal split |

Reduction levers, ranked by impact

Materials

Highest impact

- Shift polyester to recycled PET
- Source low-carbon cotton
- Engage top suppliers on energy and fertilizer

Logistics

Medium

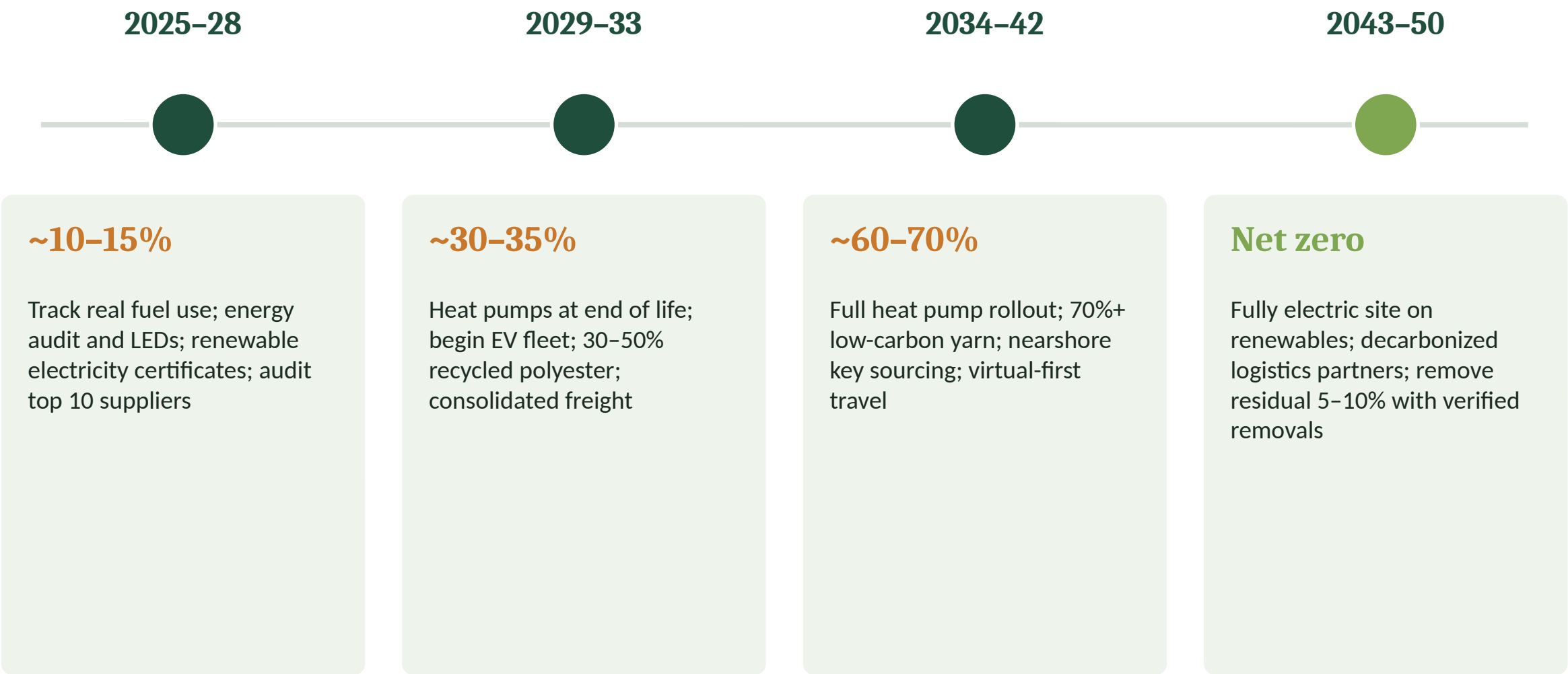
- Consolidate shipments
- Use carriers with low-emission fleets
- Nearshore where feasible

Operations

Lower, but direct control

- Heat pumps as gas equipment retires
- Electric forklifts and vehicles
- Energy management and LED retrofits

A phased path to net zero by 2050



What this project shows

- GHG Protocol boundary setting and Scope 1, 2 and 3 accounting for an SME
- Turning incomplete client data into a documented, auditable estimate
- Translating results into a roadmap a small business can act on

Company name and all figures are fictional. Original client data is confidential.